Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/28/2006
PI Name:	Karsenty, Gerard M.D., Ph.D.		
Project Title:	Leptin as a regulator of bone formation in microgravity		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Bone Loss Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	77030	**Congressional District:**	25
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NSBRI
Start Date:	02/01/2001	**End Date:**	05/31/2004
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	BL.002.07		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Leptin inhibits bone formation through unknown mechanisms following binding to hypothalamic receptors. We show here that unlike body weight bone formation cannot be affected by modulating plasma leptin levels. In contrast, when delivered centrally, minimal amounts of leptin that do not affect body weight decreasees bone mass by acting on a distinct group of hypothalamic neurons forming the central center of bone formation (CCBF). Thus, leptin regulation of bone formation does not follow the hormonal paradigm that applies for body weight control.		
Rationale for HRP Directed Research:			

Research Impact/Earth Benefits:	Leptin is a polypeptide synthesized primarily by the adipocytes that act as the afferent signal in a negative regulatory loop controlling body weight. This negative regulatory loop includes transport of leptin through the blood brain barrier to the hypothalamus, its binding to specific receptors on neurons located primarily in the arcuate and ventromedial hypothalamic (VMH) nuclei, followed by the release of bone formation (CCBF) whose function is regulated by leptin. Finally, we show that the efferent signal of the CCBF is also of neuronal origin. Together these observations reveal that leptin does not follow the paradigm of a hormonal mode of action to control bone mass. Moreover, they propose a novel bone formation-enhancing treatment for osteoporosis.
Task Progress:	No progress report this period.
Bibliography Type:	Description: (Last Updated: 02/27/2004)